

# Westchester County Data Platform

## Methodology Report

Arcanum Research Initiative

October 2025

### Contents

# 1 Executive Summary

**Project Purpose:** The Westchester County Data Platform provides comprehensive analysis of demographic, economic, transit, and municipal data for Westchester County, New York through integrated open data sources, interactive mapping, and professional-grade dashboards.

**Methodology Overview:** Multi-source data collection from U.S. Census Bureau, Metro-North Railroad GTFS feeds, and NY State Open Data portal, processed into Druck-compliant Excel files and interactive web visualizations.

**Key Findings:** Successfully integrated 56 Metro-North stations, 241 census tracts, and county-level demographics for comprehensive geographic and statistical analysis.

**Deliverables:** 4 Excel datasets, interactive React/TypeScript web platform, FastAPI backend, and comprehensive documentation.

## 2 Research Objectives

### 2.1 Primary Research Questions

- What are the current demographic characteristics of Westchester County?
- How accessible is public transit (Metro-North) across the county?
- What geographic and economic patterns exist across municipalities?
- How can open data sources be integrated for evidence-based policy analysis?

### 2.2 Success Criteria

- Complete data collection from all primary sources (Census, GTFS, NY State)
- Generation of Druck-compliant Excel files (ONE sheet per file)
- Operational web platform with interactive dashboards
- Professional documentation suitable for government stakeholders

## 3 Methodology

### 3.1 Data Sources

- **U.S. Census Bureau API:** American Community Survey (ACS) 5-Year Estimates (2022)
  - County-level demographics (population, income, housing, employment)
  - Census tract-level data (241 tracts)
  - Municipality-level aggregations (1,285 places in NY State, filtered to Westchester)
  - API endpoint: <https://api.census.gov/data/2022/acs/acs5>
  - No API key required (recommended for higher rate limits)
- **Metro-North Railroad GTFS:** General Transit Feed Specification
  - Station locations, routes, and accessibility data

- Downloaded from: <http://web.mta.info/developers/data/mnr/google-transit.zip>
- Geographic filtering: Lat 40.9-41.4, Lon -74.0 to -73.5
- Result: 56 stations in Westchester County

- **NY State Open Data (Socrata):** data.ny.gov platform

- Crime statistics (1,611 records for Westchester)
- Health facilities (330 facilities)
- Property assessment data (pending integration)
- App token recommended for API access

### 3.2 Analytical Framework

**Core Approach:** Geographic Information Systems (GIS) analysis combined with statistical demographic analysis.

**Transit Accessibility Model:**

$$\text{Coverage Area} = \pi r^2 \times N_{\text{stations}} \quad (1)$$

where  $r = 1$  mile (walkable radius) and  $N_{\text{stations}} = 56$

**Implementation Steps:**

1. Data Collection: Automated API requests to Census, GTFS download, Socrata queries
2. Geographic Filtering: Bounding box filter for Westchester County coordinates
3. Data Transformation: JSON/CSV conversion, column standardization
4. Excel Generation: Druck-compliant formatting (one sheet per file)
5. Visualization: React/TypeScript dashboards with Recharts and Leaflet mapping

### 3.3 Quality Assurance

**Validation Procedures:**

- Automated Excel validation (ONE sheet per file requirement)
- Geographic coordinate verification (within Westchester bounds)
- Data completeness checks (non-null critical fields)
- Cross-reference with official county statistics

**Accuracy Metrics:** 100% compliance with Druck standards, 0 critical violations in Excel validation.

## 4 Implementation Details

### 4.1 Software and Tools

- **Primary Analysis:** Python 3.11+ with pandas, geopandas, requests
- **Data Processing:** Custom processors for demographics and transit
- **Excel Generation:** openpyxl with Druck formatting standards
- **Backend API:** FastAPI with 8 operational endpoints
- **Frontend:** React 18, TypeScript, Vite, Tailwind CSS
- **Mapping:** Leaflet for interactive geographic visualization
- **Charts:** Recharts for statistical visualizations

### 4.2 Technical Architecture

- **Data Layer:** Raw data in `Technical/data/raw/`, processed in `Technical/data/processed/`
- **API Layer:** FastAPI REST endpoints serving JSON data
- **Frontend Layer:** React SPA with client-side routing
- **Output Layer:** Excel files in `Output/Data/Results/`, PDFs in `Output/PDFs/`

## 5 Data Processing Pipeline

### 5.1 Step 1: Data Acquisition

```
python Technical/scripts/download_all_data.py
```

Downloads all data sources, handles API authentication, geographic filtering

### 5.2 Step 2: Excel Generation

```
python Technical/src/processors/generate_all_excel.py
```

Processes raw data into Druck-compliant Excel files with proper formatting

### 5.3 Step 3: Validation

```
python Technical/scripts/validate_excel.py
```

Verifies ONE sheet per file, validates column names, checks file integrity

## 6 Limitations and Future Work

### 6.1 Current Limitations

- Municipality filtering requires refinement (captured 1,285 NY places, needs Westchester-specific list)
- Property tax data partially integrated (parcel-level analysis pending)
- Historical trend analysis limited to single-year snapshot (2022)
- Transit ridership data not included in current GTFS import

### 6.2 Future Enhancements

- Multi-year time series analysis
- Advanced transit coverage heat mapping
- Integration with county budget systems
- Real-time data updates via scheduled jobs
- PostgreSQL + PostGIS for spatial queries

## 7 References

- U.S. Census Bureau. (2022). *American Community Survey 5-Year Estimates*. Retrieved from <https://www.census.gov/programs-surveys/acs>
- Metropolitan Transportation Authority. (2025). *Metro-North Railroad GTFS Data*. Retrieved from <http://web.mta.info/developers/>
- New York State. (2025). *Open Data Portal*. Retrieved from <https://data.ny.gov>
- Westchester County GIS. (2025). *Tax Parcel Data*. Retrieved from Westchester County Open Data

## 8 Appendices

### 8.1 Appendix A: API Endpoints

- GET `/api/municipalities` - List of county municipalities
- GET `/api/transit/stations` - Metro-North station data
- GET `/api/demographics/county` - County-level demographics
- GET `/api/demographics/tracts` - Census tract data
- GET `/api/demographics/municipalities` - Municipality demographics
- GET `/api/stats` - Platform statistics and data availability

## 8.2 Appendix B: Excel File Inventory

1. [2025.10.13] westchester\_county\_demographics.xlsx
2. [2025.10.13] westchester\_census\_tracts.xlsx
3. [2025.10.13] westchester\_metro\_north\_stations.xlsx
4. [2025.10.13] westchester\_transit\_accessibility.xlsx

## 8.3 Appendix C: Census Variables

| Variable Code | Description             |
|---------------|-------------------------|
| B01003_001E   | Total Population        |
| B01002_001E   | Median Age              |
| B19013_001E   | Median Household Income |
| B25077_001E   | Median Home Value       |
| B23025_005E   | Unemployed Population   |

Table 1: Key Census ACS Variables Used